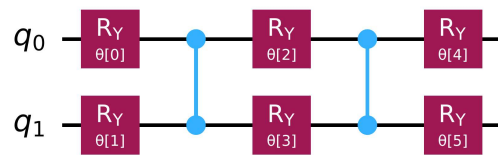


RISULTATI

VQE – Preparazione dello stato parametrizzato $|\psi(\theta)\rangle$ - Ansatz

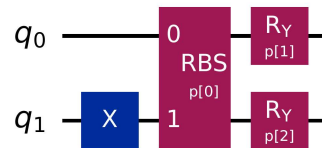
Circuito Quantistico - Dimero

Tre circuiti $U(\theta)$, che a partire da $|00\rangle$ preparano $|\psi(\theta)\rangle = U(\theta)|00\rangle$, confrontati sulla fedeltà $F = |\langle\psi(\theta)|\psi_0\rangle|^2$ rispetto allo stato fondamentale vero



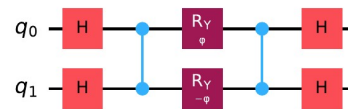
HA — 6 parametri

HA: termine di confronto (baseline generica, senza usare la fisica del problema)

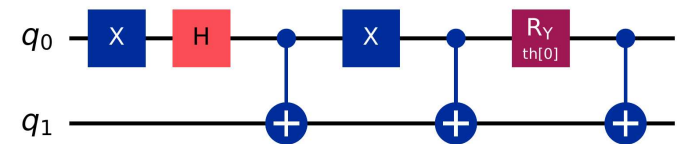


PMA esteso — 3 parametri

$RBS(\varphi) = 2 \text{ CZ} + 4 \text{ H}$

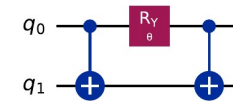


$RBS(\varphi) = e^{2i\varphi(s_1^x s_2^y - s_1^y s_2^x)}$ - rotazione reale
- Reconfigurable Beam Splitter



PMA base — 1 parametro (2 rami, $b/J \lesssim 2$)

$W(\theta) = 2 \text{ CX}$



$W(\theta) = e^{-2i\theta s_1^x s_2^y}$ - rotazione reale